AIRCRAFT MAINTENANCE MANUAL (AMM)

BOEING 747-8F FLEET

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VOLUME 2: REMOVAL, INSTALLATION, AND TESTING PROCEDURES

INTRODUCTION AND SAFETY PROTOCOLS

This Aircraft Maintenance Manual (AMM) provides the authorized procedures for removing, installing, and testing components on the 747-8F fleet.

General Safety Warning: Always ensure hydraulic systems are depressurized and electrical busses are isolated (Lockout/Tagout) prior to commencing any removal procedures. Prior to executing this procedure, technicians must verify the operational status of the interfacing subsystems. Inspect all primary load-bearing surfaces for signs of fatigue, stress-corrosion cracking, or delamination. If superficial damage exceeds the limits specified in the Structural Repair Manual (SRM), an Engineering Liaison (EL) team must be consulted before proceeding. All electrical connectors mating to this component must be checked for bent pins, dielectric grease degradation, and proper locking wire installation. Ensure that the localized work area is completely free of Foreign Object Debris (FOD) prior to closing any access panels.

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1 CHAPTER 01 - STANDARD PRACTICES

1.1 01-10-00: GENERAL SAFETY

WARNING: BEFORE STARTING MAINTENANCE, MAKE SURE THAT THE AIRCRAFT IS SAFELY GROUNDED.

Prior to executing this procedure, technicians must verify the operational status of the interfacing subsystems. Inspect all primary load-bearing surfaces for signs of fatigue, stress-corrosion cracking, or delamination. If superficial damage exceeds the limits specified in the Structural Repair Manual (SRM), an Engineering Liaison (EL) team must be consulted before proceeding. All electrical connectors mating to this component must be checked for bent pins, dielectric grease degradation, and proper locking wire installation. Ensure that the localized work area is completely free of Foreign Object Debris (FOD) prior to closing any access panels.

Routine servicing intervals for this assembly are dictated by the Maintenance Review Board Report (MRBR) and the localized operator's approved maintenance program. When performing functional checks, utilize calibrated ground support equipment (GSE) and verify that the calibration dates are active. Do not exceed the maximum hydraulic test pressure of 3,000 PSI unless explicitly instructed by a subsequent test protocol. In the event of a system fault during the return-to-service test, interrogate the Central Maintenance Computing System (CMCS) for isolated fault codes and refer to the Fault Isolation Manual (FIM).

Lockout and Tagout Procedures

When working on high-voltage electrical systems or high-pressure hydraulic lines, technicians must adhere to strict Lockout/Tagout (LOTO) protocols. Place a visible warning tag on the flight deck control panels indicating that maintenance is in progress. Ensure the relevant circuit breakers are pulled and collared.

During removal, cap and plug all open fluid lines immediately to prevent contamination of the hydraulic or fuel manifolds. Use Skydrol-resistant caps for hydraulic lines. Do not allow fluids to pool in the lower lobe bilges. Clean any spilled fluids using an approved solvent (e.g., MEK or isopropyl alcohol) and lint-free cloths. When installing the replacement unit, observe all torque specifications listed in Appendix A. Safety-wire all turnbuckles, fluid fittings, and cannon plugs in accordance with Standard Wiring Practices Manual (SWPM) guidelines.

Foreign Object Debris (FOD) Prevention

FOD represents a critical threat to aircraft safety. Prior to executing this procedure, technicians must verify the operational status of the interfacing subsystems. Inspect all primary load-bearing surfaces for signs of fatigue, stress-corrosion cracking, or delamination. If superficial damage exceeds the limits specified in the Structural Repair Manual (SRM), an Engineering Liaison (EL) team must be consulted before proceeding. All electrical connectors mating to this component must be checked for bent pins, dielectric grease degradation, and proper locking wire installation. Ensure that the localized work area is completely free of Foreign Object Debris (FOD) prior to closing any access panels.All tools must be accounted for before and after shifting shifts. Use shadowed toolboxes to ensure rapid visual verification of tool inventory.

2 CHAPTER 12 - SERVICING

2.1 12-10-00: FLUID REPLENISHMENT

Routine servicing intervals for this assembly are dictated by the Maintenance Review Board Report (MRBR) and the localized operator's approved maintenance program. When performing functional checks, utilize calibrated ground support equipment (GSE) and verify that the calibration dates are active. Do not exceed the maximum hydraulic test pressure of 3,000 PSI unless explicitly instructed by a subsequent test protocol. In the event of a system fault during the return-to-service test, interrogate the Central Maintenance Computing System (CMCS) for isolated fault codes and refer to the Fault Isolation Manual (FIM).

Hydraulic Fluid Servicing

The 747-8F utilizes Skydrol LD-4 or equivalent Type IV phosphate ester hydraulic fluid. During removal, cap and plug all open fluid lines immediately to prevent contamination of the hydraulic or fuel manifolds. Use Skydrol-resistant caps for hydraulic lines. Do not allow fluids to pool in the lower lobe bilges. Clean any spilled fluids using an approved solvent (e.g., MEK or isopropyl alcohol) and lint-free cloths. When installing the replacement unit, observe all torque specifications listed in Appendix A. Safety-wire all turnbuckles, fluid fittings, and cannon plugs in accordance with Standard Wiring Practices Manual (SWPM) guidelines.

Ensure the aircraft is parked on level ground before taking fluid level readings from the reservoir sight gauges. The hydraulic reservoirs for Systems 1, 2, 3, and 4 must be pressurized with bleed air to prevent pump cavitation during startup. If reservoir pressure drops below 35 PSI, service the pneumatic head using a nitrogen cart.

Prior to executing this procedure, technicians must verify the operational status of the interfacing subsystems. Inspect all primary load-bearing surfaces for signs of fatigue, stress-corrosion cracking, or delamination. If superficial damage exceeds the limits specified in the Structural Repair Manual (SRM), an Engineering Liaison (EL) team must be consulted before proceeding. All electrical connectors mating to this component must be checked for bent pins, dielectric grease degradation, and proper locking wire installation. Ensure that the localized work area is completely free of Foreign Object Debris (FOD) prior to closing any access panels.

Engine Oil Servicing

Service the GEnx-2B engine oil tanks within 30 to 60 minutes after engine shutdown to ensure accurate readings. Do not mix different brands of synthetic turbine oil (MIL-PRF-23699) unless explicitly approved by the Engineering Liaison team. Routine servicing intervals for this assembly are dictated by the Maintenance Review Board Report (MRBR) and the localized operator's approved maintenance program. When performing functional checks, utilize calibrated ground support equipment (GSE) and verify that the calibration dates are active. Do not exceed the maximum hydraulic test pressure of 3,000 PSI unless explicitly instructed by a subsequent test protocol. In the event of a system fault during the return-to-service test, interrogate the Central Maintenance Computing System (CMCS) for isolated fault codes and refer to the Fault Isolation Manual (FIM).

3 CHAPTER 21 - AIR CONDITIONING

3.1 21-50-01-400: AIR CYCLE MACHINE PACK VALVE

Prior to executing this procedure, technicians must verify the operational status of the interfacing subsystems. Inspect all primary load-bearing surfaces for signs of fatigue, stress-corrosion cracking, or delamination. If superficial damage exceeds the limits specified in the Structural Repair Manual (SRM), an Engineering Liaison (EL) team must be consulted before proceeding. All electrical connectors mating to this component must be checked for bent pins, dielectric grease degradation, and proper locking wire installation. Ensure that the localized work area is completely free of Foreign Object Debris (FOD) prior to closing any access panels.

DIAGRAM: PACK VALVE
V-Band clamp torque sequence.

Figure 1: Pack Valve Installation

Procedure Steps:

1. Ensure bleed air pressure is completely exhausted from the pneumatic manifold. During removal, cap and plug all open fluid lines immediately to prevent contamination of the hydraulic or fuel manifolds. Use Skydrol-resistant caps for hydraulic lines. Do not allow fluids to pool in the lower lobe bilges. Clean any spilled fluids using an approved solvent (e.g., MEK or isopropyl alcohol) and lint-free cloths. When installing the replacement unit, observe all torque specifications listed in Appendix A. Safety-wire all turnbuckles, fluid fittings, and cannon plugs in accordance with Standard Wiring Practices Manual (SWPM) guidelines.
2. Install a new O-ring seal (SL-990-1) onto the valve mating flange.
3. Position the Pack Valve (VLV-ACM-990 or approved alternate) onto the duct flange.
4. Install the V-Band clamp.
5. Torque the V-Band clamp nut to 75 in-lbs. Tap the circumference of the clamp with a non-metallic mallet, then re-torque to 75 in-lbs.
6. Connect the electrical connector to the valve actuator mechanism.
7. **NOTE:** If installing the True Alternate part (**VLV-ACM-991-ALT**), mechanics should connect the maintenance laptop to the CMCS and verify the Pack Controller software version.

Routine servicing intervals for this assembly are dictated by the Maintenance Review Board Report (MRBR) and the localized operator's approved maintenance program. When performing functional checks, utilize calibrated ground support equipment (GSE) and verify that the calibration dates are active. Do not exceed the maximum hydraulic test pressure of 3,000 PSI unless explicitly instructed by a subsequent test protocol. In the event of a system fault during the return-to-service test, interrogate the Central Maintenance Computing System (CMCS) for isolated fault codes and refer to the Fault Isolation Manual (FIM).

4 CHAPTER 24 - ELECTRICAL POWER

4.1 24-20-00: AC GENERATION SYSTEM

Prior to executing this procedure, technicians must verify the operational status of the interfacing subsystems. Inspect all primary load-bearing surfaces for signs of fatigue, stress-corrosion cracking, or delamination. If superficial damage exceeds the limits specified in the Structural Repair Manual (SRM), an Engineering Liaison (EL) team must be consulted before proceeding. All electrical connectors mating to this component must be checked for bent pins, dielectric grease degradation, and proper locking wire installation. Ensure that the localized work area is completely free of Foreign Object Debris (FOD) prior to closing any access panels.

Integrated Drive Generator (IDG) Servicing

The IDG combines a constant speed drive (CSD) and a generator into a single unit. Ensure the IDG oil cooler is free of obstructions. During removal, cap and plug all open fluid lines immediately to prevent contamination of the hydraulic or fuel manifolds. Use Skydrol-resistant caps for hydraulic lines. Do not allow fluids to pool in the lower lobe bilges. Clean any spilled fluids using an approved solvent (e.g., MEK or isopropyl alcohol) and lint-free cloths. When installing the replacement unit, observe all torque specifications listed in Appendix A. Safety-wire all turnbuckles, fluid fittings, and cannon plugs in accordance with Standard Wiring Practices Manual (SWPM) guidelines.

Check the IDG oil level sight glass. If the oil level is low, replenish using the pressure fill port to avoid introducing air bubbles into the system. Routine servicing intervals for this assembly are dictated by the Maintenance Review Board Report (MRBR) and the localized operator's approved maintenance program. When performing functional checks, utilize calibrated ground support equipment (GSE) and verify that the calibration dates are active. Do not exceed the maximum hydraulic test pressure of 3,000 PSI unless explicitly instructed by a subsequent test protocol. In the event of a system fault during the return-to-service test, interrogate the Central Maintenance Computing System (CMCS) for isolated fault codes and refer to the Fault Isolation Manual (FIM).

Standby Power System

The standby power system consists of the main battery, auxiliary battery, and static inverters. Prior to executing this procedure, technicians must verify the operational status of the interfacing subsystems. Inspect all primary load-bearing surfaces for signs of fatigue, stress-corrosion cracking, or delamination. If superficial damage exceeds the limits specified in the Structural Repair Manual (SRM), an Engineering Liaison (EL) team must be consulted before proceeding. All electrical connectors mating to this component must be checked for bent pins, dielectric grease degradation, and proper locking wire installation. Ensure that the localized work area is completely free of Foreign Object Debris (FOD) prior to closing any access panels.During a complete loss of AC power, the standby system provides a minimum of 30 minutes of emergency power to essential flight instruments.

5 CHAPTER 27 - FLIGHT CONTROLS

5.1 27-11-05-400: INBOARD AILERON PCU

WARNING: KEEP PERSONS AND EQUIPMENT CLEAR OF ALL FLIGHT CONTROL SURFACES WHEN HYDRAULIC POWER IS APPLIED.

During removal, cap and plug all open fluid lines immediately to prevent contamination of the hydraulic or fuel manifolds. Use Skydrol-resistant caps for hydraulic lines. Do not allow fluids to pool in the lower lobe bilges. Clean any spilled fluids using an approved solvent (e.g., MEK or isopropyl alcohol) and lint-free cloths. When installing the replacement unit, observe all torque specifications listed in Appendix A. Safety-wire all turnbuckles, fluid fittings, and cannon plugs in accordance with Standard Wiring Practices Manual (SWPM) guidelines.

Procedure Steps:

1. Place the Inboard Aileron Power Control Unit (PCU) into the mounting bracket. Prior to executing this procedure, technicians must verify the operational status of the interfacing subsystems. Inspect all primary load-bearing surfaces for signs of fatigue, stress-corrosion cracking, or delamination. If superficial damage exceeds the limits specified in the Structural Repair Manual (SRM), an Engineering Liaison (EL) team must be consulted before proceeding. All electrical connectors mating to this component must be checked for bent pins, dielectric grease degradation, and proper locking wire installation. Ensure that the localized work area is completely free of Foreign Object Debris (FOD) prior to closing any access panels.
2. **NOTE FOR LUG WEAR:** If inspection reveals attach lug wear exceeding 0.050 inches, the oversized PCU (**PCU-AIL-INB-03-OS**) must be utilized. The attach lug must be re-reamed to accept the oversized rod-end bearing prior to installation.
3. Install the mounting bolts and torque to standard values (See Appendix A).
4. Connect hydraulic lines for System 1, 2, and 3 to the respective manifold ports on the PCU.
5. Perform the Hydraulic System Bleeding Procedure to remove entrapped air from the lines.
6. Execute the Flight Control Rigging check using the onboard maintenance system to assure null bias is correctly calibrated.

Routine servicing intervals for this assembly are dictated by the Maintenance Review Board Report (MRBR) and the localized operator's approved maintenance program. When performing functional checks, utilize calibrated ground support equipment (GSE) and verify that the calibration dates are active. Do not exceed the maximum hydraulic test pressure of 3,000 PSI unless explicitly instructed by a subsequent test protocol. In the event of a system fault during the return-to-service test, interrogate the Central Maintenance Computing System (CMCS) for isolated fault codes and refer to the Fault Isolation Manual (FIM).

6 CHAPTER 28 - FUEL SYSTEMS

6.1 28-22-10-000: CENTER WING TANK BOOST PUMP

DANGER: FUEL VAPORS ARE HIGHLY FLAMMABLE AND TOXIC. STRICT CONFINED SPACE ENTRY PROCEDURES MUST BE FOLLOWED.

Prior to executing this procedure, technicians must verify the operational status of the interfacing subsystems. Inspect all primary load-bearing surfaces for signs of fatigue, stress-corrosion cracking, or delamination. If superficial damage exceeds the limits specified in the Structural Repair Manual (SRM), an Engineering Liaison (EL) team must be consulted before proceeding. All electrical connectors mating to this component must be checked for bent pins, dielectric grease degradation, and proper locking wire installation. Ensure that the localized work area is completely free of Foreign Object Debris (FOD) prior to closing any access panels.

DIAGRAM: TANK ENTRY
Purging and ventilation hookup.

Figure 2: CWT Access

Procedure Steps:

1. Defuel the Center Wing Tank entirely.
2. Connect purging equipment and force-ventilate the tank for a minimum of 2 hours. During removal, cap and plug all open fluid lines immediately to prevent contamination of the hydraulic or fuel manifolds. Use Skydrol-resistant caps for hydraulic lines. Do not allow fluids to pool in the lower lobe bilges. Clean any spilled fluids using an approved solvent (e.g., MEK or isopropyl alcohol) and lint-free cloths. When installing the replacement unit, observe all torque specifications listed in Appendix A. Safety-wire all turnbuckles, fluid fittings, and cannon plugs in accordance with Standard Wiring Practices Manual (SWPM) guidelines.
3. Verify atmospheric oxygen levels are above 19.5% and LEL is below 10% using a calibrated multi-gas meter.
4. Disconnect the 3-phase electrical connector from the pump housing (PMP-CWT-88A).
5. Remove the mounting ring bolts and extract the boost pump assembly.
6. Discard the crush gasket (GSK-CWT-1). A new gasket must be used during installation.

Routine servicing intervals for this assembly are dictated by the Maintenance Review Board Report (MRBR) and the localized operator's approved maintenance program. When performing functional checks, utilize calibrated ground support equipment (GSE) and verify that the calibration dates are active. Do not exceed the maximum hydraulic test pressure of 3,000 PSI unless explicitly instructed by a subsequent test protocol. In the event of a system fault during the return-to-service test, interrogate the Central Maintenance Computing System (CMCS) for isolated fault codes and refer to the Fault Isolation Manual (FIM).

7 CHAPTER 29 - HYDRAULIC POWER

7.1 29-11-00: ENGINE DRIVEN PUMPS (EDP)

Prior to executing this procedure, technicians must verify the operational status of the interfacing subsystems. Inspect all primary load-bearing surfaces for signs of fatigue, stress-corrosion cracking, or delamination. If superficial damage exceeds the limits specified in the Structural Repair Manual (SRM), an Engineering Liaison (EL) team must be consulted before proceeding. All electrical connectors mating to this component must be checked for bent pins, dielectric grease degradation, and proper locking wire installation. Ensure that the localized work area is completely free of Foreign Object Debris (FOD) prior to closing any access panels.

EDP Removal Precautions

The Engine Driven Pumps provide the primary 3,000 PSI hydraulic pressure for the aircraft. Before removing an EDP, you must close the respective hydraulic shutoff valve located in the engine strut. During removal, cap and plug all open fluid lines immediately to prevent contamination of the hydraulic or fuel manifolds. Use Skydrol-resistant caps for hydraulic lines. Do not allow fluids to pool in the lower lobe bilges. Clean any spilled fluids using an approved solvent (e.g., MEK or isopropyl alcohol) and lint-free cloths. When installing the replacement unit, observe all torque specifications listed in Appendix A. Safety-wire all turnbuckles, fluid fittings, and cannon plugs in accordance with Standard Wiring Practices Manual (SWPM) guidelines.

System Bleeding

After installing a new EDP, the hydraulic system must be bled to remove air. Connect a hydraulic ground cart to the return module and circulate fluid for a minimum of 15 minutes. Routine servicing intervals for this assembly are dictated by the Maintenance Review Board Report (MRBR) and the localized operator's approved maintenance program. When performing functional checks, utilize calibrated ground support equipment (GSE) and verify that the calibration dates are active. Do not exceed the maximum hydraulic test pressure of 3,000 PSI unless explicitly instructed by a subsequent test protocol. In the event of a system fault during the return-to-service test, interrogate the Central Maintenance Computing System (CMCS) for isolated fault codes and refer to the Fault Isolation Manual (FIM).Cycle the flight controls (rudder, elevators, ailerons) through their full range of motion at least three times to purge air from the actuator lines.

8 CHAPTER 32 - LANDING GEAR

8.1 32-30-01-000: MLG RETRACTION ACTUATOR - REMOVAL

WARNING: MAKE SURE THE LANDING GEAR DOWNLOCK PINS ARE INSTALLED BEFORE YOU START THIS PROCEDURE.

During removal, cap and plug all open fluid lines immediately to prevent contamination of the hydraulic or fuel manifolds. Use Skydrol-resistant caps for hydraulic lines. Do not allow fluids to pool in the lower lobe bilges. Clean any spilled fluids using an approved solvent (e.g., MEK or isopropyl alcohol) and lint-free cloths. When installing the replacement unit, observe all torque specifications listed in Appendix A. Safety-wire all turnbuckles, fluid fittings, and cannon plugs in accordance with Standard Wiring Practices Manual (SWPM) guidelines.

Procedure Steps:

1. Supply electrical power to the aircraft.
2. Depressurize the Right Hydraulic System.
3. Position an equipment hoist above the MLG wheel well.
4. Attach the hoisting sling to the lifting eyes on the actuator body.
5. Disconnect the hydraulic supply and return lines from the actuator manifold.
6. Remove the upper and lower trunnion attach bolts.
7. Carefully lower the actuator assembly.

8.2 32-30-01-400: MLG RETRACTION ACTUATOR - INSTALLATION

CAUTION: REFER TO THE ILLUSTRATED PARTS CATALOG (IPC 32-30-01) TO VERIFY THE PART NUMBER.

Procedure Steps:

1. Apply Boeing standard grease (MIL-PRF-81322) to the upper and lower attach bolts. Prior to executing this procedure, technicians must verify the operational status of the interfacing subsystems. Inspect all primary load-bearing surfaces for signs of fatigue, stress-corrosion cracking, or delamination. If superficial damage exceeds the limits specified in the Structural Repair Manual (SRM), an Engineering Liaison (EL) team must be consulted before proceeding. All electrical connectors mating to this component must be checked for bent pins, dielectric grease degradation, and proper locking wire installation. Ensure that the localized work area is completely free of Foreign Object Debris (FOD) prior to closing any access panels.
2. **NOTE FOR OVERSIZED PARTS:** If installing the oversized alternate actuator (**ACT-747-OVR**), you **must** first install structural modification bracket **B-99** using the titanium hardware kit **HK-22**. Torque all titanium bolts to 450 in-lbs.
3. Install the upper trunnion attach bolt, washer, and nut. Torque standard nuts to 320 in-lbs.
4. Install the lower attach bolt. Torque to 320 in-lbs.
5. Pressurize the Right Hydraulic System and inspect all connections for leaks. Routine servicing intervals for this assembly are dictated by the Maintenance Review Board Report (MRBR) and the localized operator's approved maintenance program. When performing functional checks, utilize calibrated ground support equipment (GSE) and verify that the calibration dates are active. Do not exceed the maximum hydraulic test pressure of 3,000 PSI unless explicitly instructed by a subsequent test protocol. In the event of a system fault during the return-to-service test, interrogate the Central Maintenance Computing System (CMCS) for isolated fault codes and refer to the Fault Isolation Manual (FIM).

9 CHAPTER 49 - APU

9.1 49-10-22-400: APU STARTER GENERATOR

CAUTION: THE STARTER GENERATOR WEIGHS APPROXIMATELY 85 LBS (38 KG). USE LIFTING EQUIPMENT.

Prior to executing this procedure, technicians must verify the operational status of the interfacing subsystems. Inspect all primary load-bearing surfaces for signs of fatigue, stress-corrosion cracking, or delamination. If superficial damage exceeds the limits specified in the Structural Repair Manual (SRM), an Engineering Liaison (EL) team must be consulted before proceeding. All electrical connectors mating to this component must be checked for bent pins, dielectric grease degradation, and proper locking wire installation. Ensure that the localized work area is completely free of Foreign Object Debris (FOD) prior to closing any access panels.

Procedure Steps:

1. Inspect the APU accessory gearbox spline drive for damage or excessive wear. During removal, cap and plug all open fluid lines immediately to prevent contamination of the hydraulic or fuel manifolds. Use Skydrol-resistant caps for hydraulic lines. Do not allow fluids to pool in the lower lobe bilges. Clean any spilled fluids using an approved solvent (e.g., MEK or isopropyl alcohol) and lint-free cloths. When installing the replacement unit, observe all torque specifications listed in Appendix A. Safety-wire all turnbuckles, fluid fittings, and cannon plugs in accordance with Standard Wiring Practices Manual (SWPM) guidelines.
2. **NOTE FOR HD ALTERNATE:** If installing the heavy-duty 120kVA generator (**SG-APU-120kVA-ALT**), the existing aft cowl support strut must be temporarily detached. You must install the modified strut (**ST-992**) to provide adequate clearance.
3. Lubricate the generator spline drive with standard synthetic engine oil.
4. Align the generator spline with the APU accessory gearbox.
5. Install the V-Band clamp. Torque to 85 in-lbs.
6. Connect the heavy-gauge terminal lugs. Ensure proper phase alignment (A, B, C, N).

Routine servicing intervals for this assembly are dictated by the Maintenance Review Board Report (MRBR) and the localized operator's approved maintenance program. When performing functional checks, utilize calibrated ground support equipment (GSE) and verify that the calibration dates are active. Do not exceed the maximum hydraulic test pressure of 3,000 PSI unless explicitly instructed by a subsequent test protocol. In the event of a system fault during the return-to-service test, interrogate the Central Maintenance Computing System (CMCS) for isolated fault codes and refer to the Fault Isolation Manual (FIM).

10 CHAPTER 73 - ENGINE FUEL

10.1 73-11-05-400: MAIN ENGINE FUEL PUMP

Prior to executing this procedure, technicians must verify the operational status of the interfacing subsystems. Inspect all primary load-bearing surfaces for signs of fatigue, stress-corrosion cracking, or delamination. If superficial damage exceeds the limits specified in the Structural Repair Manual (SRM), an Engineering Liaison (EL) team must be consulted before proceeding. All electrical connectors mating to this component must be checked for bent pins, dielectric grease degradation, and proper locking wire installation. Ensure that the localized work area is completely free of Foreign Object Debris (FOD) prior to closing any access panels.

DIAGRAM: ACCESSORY GEARBOX
Fuel pump mounting pad.

Figure 3: GEnx-2B Mounting

Procedure Steps:

1. Verify the primary part (PMP-GENX-MFP1) or the approved true alternate (PMP-GENX-MFP2) is ready for installation.
2. Examine the Spline Drive Coupling (SPL-99). Ensure it is coated heavily with MIL-PRF-23699 engine oil.
3. **CAUTION:** Do not rotate the fuel pump impeller while it is dry. This will cause immediate damage to the internal carbon seals. During removal, cap and plug all open fluid lines immediately to prevent contamination of the hydraulic or fuel manifolds. Use Skydrol-resistant caps for hydraulic lines. Do not allow fluids to pool in the lower lobe bilges. Clean any spilled fluids using an approved solvent (e.g., MEK or isopropyl alcohol) and lint-free cloths. When installing the replacement unit, observe all torque specifications listed in Appendix A. Safety-wire all turnbuckles, fluid fittings, and cannon plugs in accordance with Standard Wiring Practices Manual (SWPM) guidelines.
4. Mate the pump to the accessory gearbox pad.
5. Install the 6 mounting stud nuts and torque to 180 in-lbs in a cross-pattern.
6. Connect the low-pressure fuel inlet line and high-pressure outlet line.

Routine servicing intervals for this assembly are dictated by the Maintenance Review Board Report (MRBR) and the localized operator's approved maintenance program. When performing functional checks, utilize calibrated ground support equipment (GSE) and verify that the calibration dates are active. Do not exceed the maximum hydraulic test pressure of 3,000 PSI unless explicitly instructed by a subsequent test protocol. In the event of a system fault during the return-to-service test, interrogate the Central Maintenance Computing System (CMCS) for isolated fault codes and refer to the Fault Isolation Manual (FIM).

11 APPENDIX A: STANDARD TORQUE LIMITS

11.1 General Specifications

Unless otherwise specified in a specific procedure within this AMM, use the following standard torque values for all hardware installations on the 747-8F airframe.

Thread Size	Torque Range (in-lbs)	Torque Range (Nm)
1/4-28 UNF	50 - 70	5.6 - 7.9
5/16-24 UNF	100 - 140	11.3 - 15.8
3/8-24 UNF	160 - 190	18.0 - 21.4
7/16-20 UNF	250 - 300	28.2 - 33.9
1/2-20 UNF	400 - 480	45.2 - 54.2

11.2 Important Safety Checklist

- **Calibration:** Ensure all torque wrenches are within their active calibration cycle prior to use. Out-of-calibration tools must be quarantined immediately.
- **Lubrication:** Standard torque limits assume ”clean and dry” threads unless the specific AMM procedure calls for lubrication (e.g., grease, anti-seize). Applying torque to wet threads without reducing the torque value will result in hardware failure.
- **Titanium Hardware:** Titanium bolts require specific torque values and MUST NOT be interchanged with standard steel hardware. Refer to specific SRM chapters for titanium usage.

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